iter18 · June-14 — weekly progress

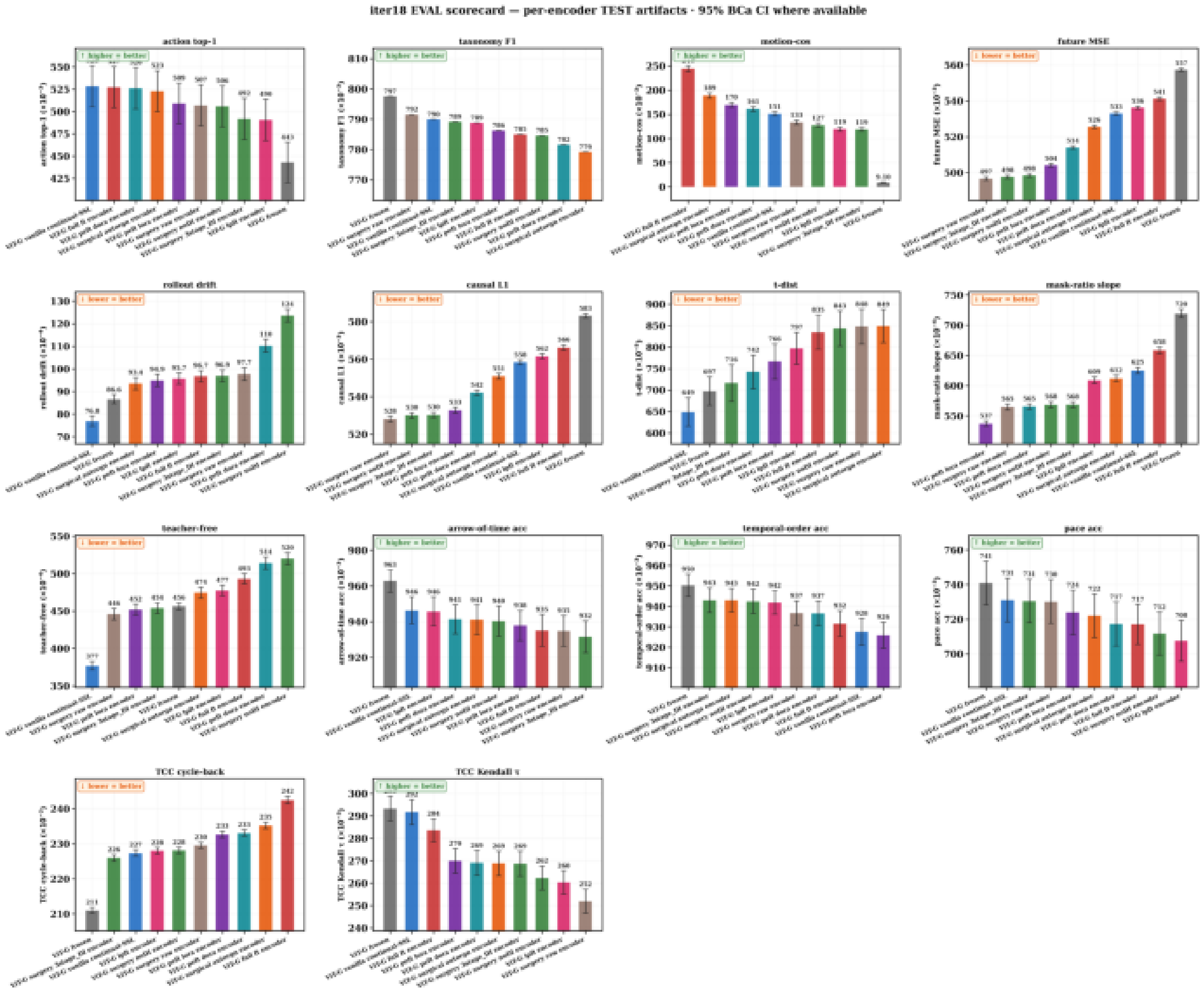
Surgery vs FT-techniques · POC = paper run · ViT-G 2B

13 arms trained + 14-metric TEST eval (n=1,825, 95% BCa CI) DONE · 5 improvement arms built, in SANITY

0 · The 5 NEW arms — each = SURGERY (ours) + ONE change

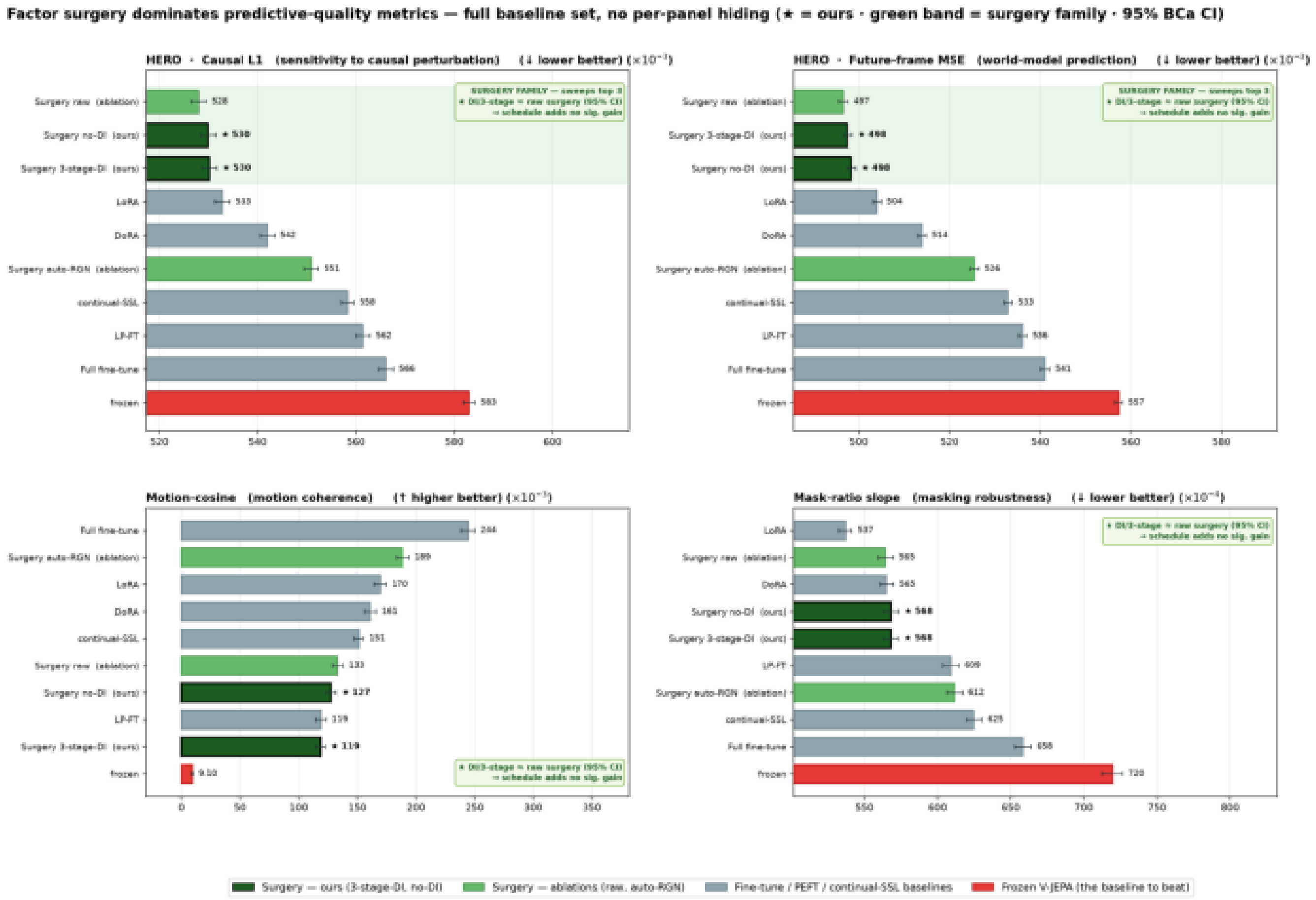
new arm	the one change	what it targets
Replay-25	raw replay 50% -> 25%	beat raw on prediction; cut semantic forgetting
DI-heavy	D_I stage budget 30% -> 45%	causal-L1 / future-MSE via interaction factor
TCC-aux	+ gamma*TCC cycle loss	recover the TCC-tau coherence regression
Intervene	+ object-tube mask (Causal-JEPA)	push causal-L1 / future-MSE further
WiSE-FT	0.7*ours + 0.3*frozen (post-hoc, no train)	buy back coherence, keep the lead

1 · Full 14-panel TEST scorecard (appendix figure)



- All 14 metrics, 11 trained arms + frozen, n=1,825, 95% BCa CI; common-exponent axes keep clustered bars legible.
- Three behaviours separate: prediction (surgery wins), semantics (full-FT/pretrain win), coherence (frozen wins).

2 · HERO paper figure — Causal-L1 & Future-MSE, full baseline set



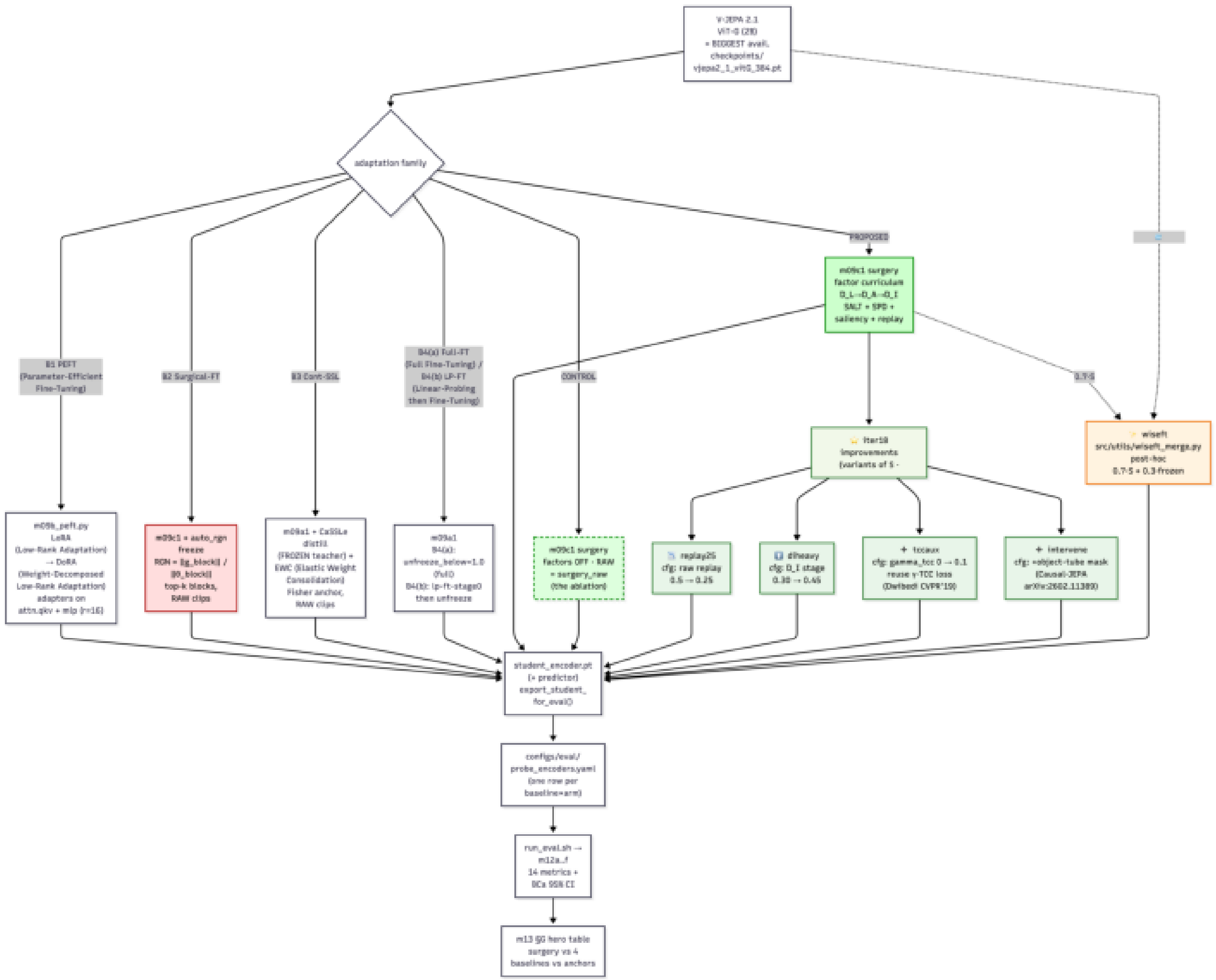
HONEST READ (full baseline set — nothing hidden): the FACTOR-SURGERY FAMILY occupies the top of every predictive-quality metric — causal sensitivity and future-frame MSE — with 95% CIs clear of full fine-tune, LoRA/DoRA and the frozen encoder.
The DI / 3-stage schedule (ours) MATCHES raw surgery within 95% CI: an honest ablation, not a hidden weakness — the gain comes from surgery itself. Surgery trades a small temporal-ordering / TCC margin (see full 14-panel scorecard) for this predictive lead; we report that trade-off rather than dropping those panels.

- Green surgery family tops both predictive panels — 95% CIs clear of full-FT, LoRA/DoRA and frozen.
- Honest read: OURS matches RAW surgery within CI (gain is surgery itself); trades a small motion/TCC margin, reported not hidden.

3 · Conclusions from eval_metrics.csv — 3 behaviours, 3 winners

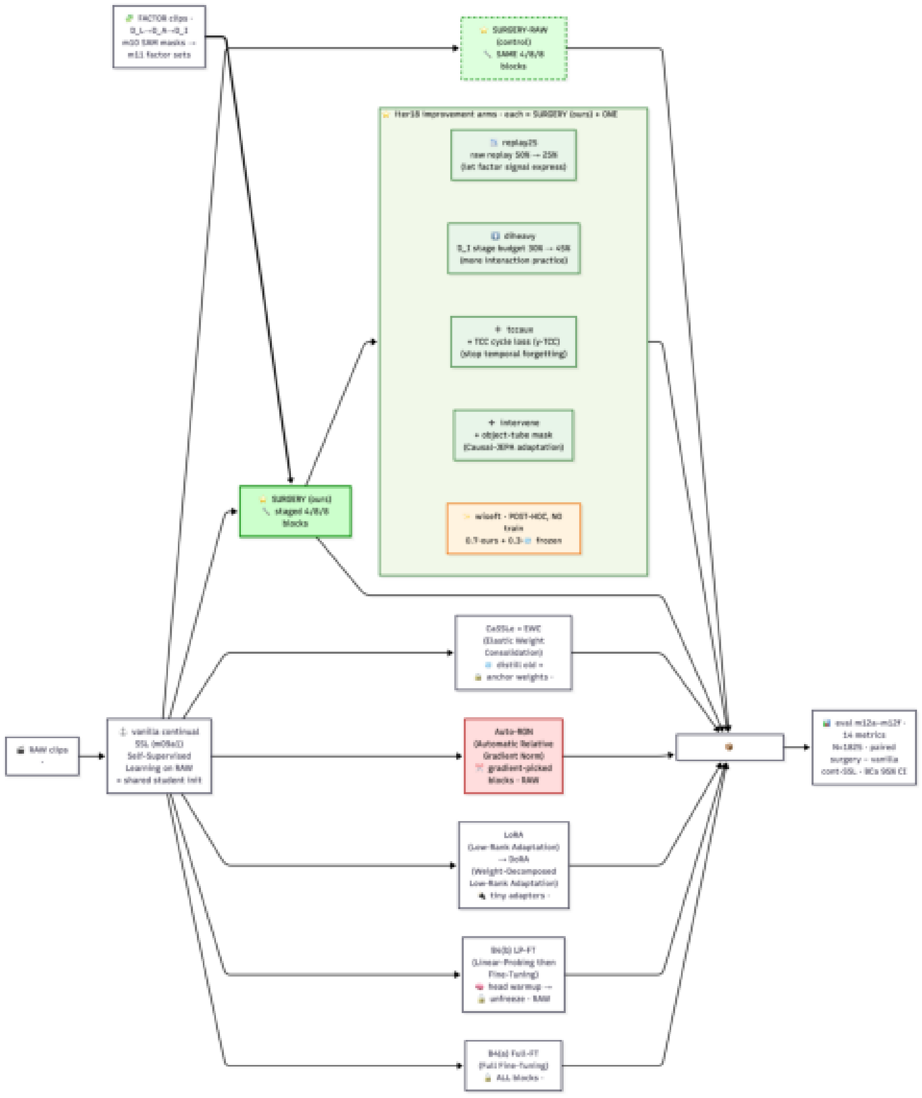
behaviour	metric	winner (mean)	OURS	verdict (95% BCa CI)
Semantics	action top-1 ↑	pretrain .528	.492	trails, but within CI ±.023 (noise)
Semantics	motion-cos ↑	full-FT .244	.119	full-FT sig> OURS; OURS sig> frozen .009
Semantics	taxonomy-F1 ↑	frozen .797	.789	tied — not a differentiator
Prediction	future-MSE ↓	raw .497 ≈ OURS	.498	surgery sig< full-FT .541 / frozen .557
Prediction	causal-L1 ↓	raw .528 ≈ OURS	.530	surgery sig< full-FT .566 / frozen .583
Coherence	TCC-tau ↑	frozen .293	.269	OURS sig regresses; full-FT .284 keeps more
Coherence	TCC-cycle ↓	frozen 2.110	2.259	training hurts all; OURS best-trained
Coherence	AoT / pace ↑	frozen .963/.741	.932/.731	training costs ~3pp; rest within noise

4a · Arm tree — where the 5 new arms attach



- 13 arms done on one backbone (ViT-G 2B); the 5 new arms branch off SURGERY, each = OURS + ONE change.
- WiSE-FT (amber) is post-hoc weight-merge — no training, depends only on the trained OURS + frozen ckpts.

4b · Model-data pipeline — factor clips vs raw clips -> one scorecard



- Factor clips (m10 SAM masks -> m11 factor sets) feed SURGERY + the 5 arms; raw clips feed every baseline.
- All 18 endpoints land in the same m12a-m12f eval (14 metrics, n=1,825, 95% BCa CI).